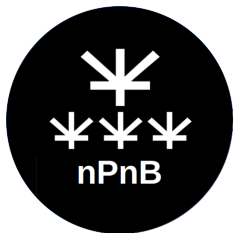


User Guide

Navigate through clustered graphs
with several scales of description

ISCPIF

May, 06, 2026



Presentation

Networks are one of the main conceptual structures for modeling complex systems, where nodes represent the basic lowest-level entities of the system and edges represent their interactions. Graph clustering algorithms are then used to identify modules : over-densely connected sets of nodes of these networks as representations of entities at higher scales of description in these systems.

bec is a C++ graph clustering program that finds the entities of a network with a hand on the **description scales** within the **nPnB framework**.

■ **nPnB** : B. GAUME, I. ACHITOUV, AND D. CHAVALARIAS : *Two antagonistic objectives for one multi-scale graph clustering framework* Nature Scientific Reports, 15 (2025), p. 13368. <https://hal.science/hal-05046050v1>

■ **bec** : B. GAUME : *Find Optimal clusterings in nPnB framework is hard, BEC.2 : an algorithm to find high-performing ones* Forthcoming (2026)

■ **description scales** : B. GAUME : *Defining the appropriate scales for describing a complex system* Forthcoming (2026)

Presentation

In the following, we describe how to navigate through graphs clustered with several scales of description.

The screenshot displays the Gaume (ISCPiF) User Guide interface, which features a central graph visualization and several interactive panels. The interface is designed for navigating through graphs clustered with several scales of description.

Informational Panels:

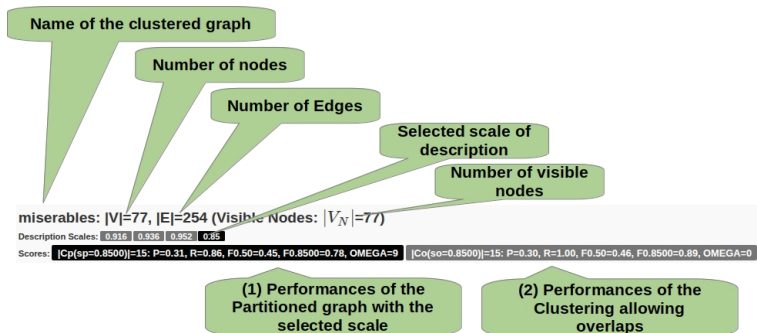
- Top Panel:** Displays graph statistics: `miserables: |V|=77, |E|=254 (Visible Nodes: |Vv|=77)`. Below this, it shows the description source: `Source: [Cytoscape-0.8.0(01)-15, P=0.31, R=0.85, F0.50=0.45, F0.5000=0.75, OMEGA=0.5]` and the cluster source: `[Cytoscape-0.8.0(01)-15, P=0.31, R=0.85, F0.50=0.45, F0.5000=0.85, OMEGA=0.5]`.
- Left Panel:** Includes a **Button Focus** and a **Button Rotation**.
- Bottom Left Panel:** A **Selects a node** callout points to a node in the graph.
- Bottom Center Panel:** A **Information on the node** callout points to a node's details.
- Top Right Panel:** A **Information on the graph and its clusterings** callout points to the top informational panel.
- Center Panel:** A **Selects Nodes-Visibility From one node** callout points to the **UPDATE NODES-VISIBILITY FROM: 01 : Gaume** panel.
- Bottom Right Panel:** A **Selects Nodes-Visibility** callout points to the **UPDATE NODES-VISIBILITY** panel.
- Right Panel:** A **Button Export** callout points to the **EXPORT** button.

Graph Visualization: The central graph shows a network of nodes and edges. Nodes are color-coded (red, green, blue) and labeled. The graph is clustered, with nodes grouped into different scales of description.

Interactive Panels:

- UPDATE NODES-VISIBILITY FROM: 01 : Gaume**: A panel for selecting nodes-visibility from one node. It includes sections for **ME**, **MY NEIGHBORS & ME**, **MY PROXIMITY (TDD)**, and **MY C₂ Module at each R₀**. Each section has a table with columns for **ADD**, **REM**, and **ON/L**.
- UPDATE NODES-VISIBILITY**: A panel for selecting nodes-visibility. It includes a **BACK (0)** button, a **ON DEGREES (deg within an INTERVAL)** section, and a **ON LABELS (lab matching with a REGEX)** section. The **ON LABELS** section includes a **lab.match()** field and a list of labels: **Equal to "lab": "lab"**, **Starting with "lab": "lab"**, **Containing "lab": "lab"**, **Ending with "lab": "lab"**, and **Not containing "lab": "lab"**. Each section has a table with columns for **ADD**, **REM**, and **ON/L**.

Informations on the graph and its clusterings



- ▶ (1) selected $\Rightarrow$ nodes are colored with C_p
- ▶ (2) selected $\Rightarrow$ nodes are colored with C_o

Let m a set of nodes, then $\omega_G(m) = \left| \{i \in \overline{m} \text{ such } \forall j \in m, \{i, j\} \in E\} \right|$

For a clustering C on a graph G , $OMEGA(C) = \sum_{m \in C} \omega_G(m)$

C is an intrinsic Clustering $\Leftrightarrow C$ is a node partition such $OMEGA(C) = 0$

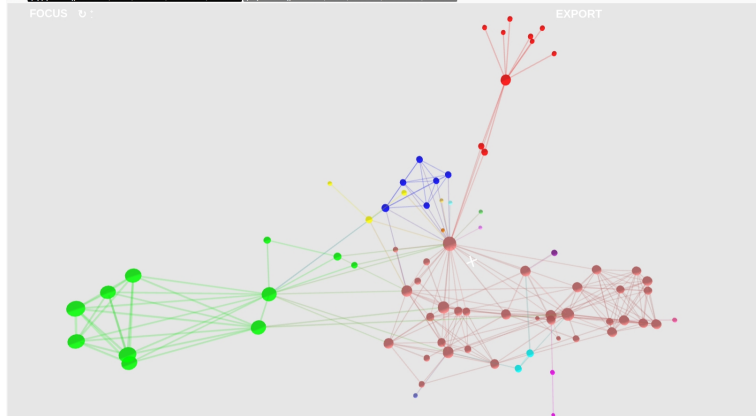
Informations on the graph and its clusterings

EXAMPLE : selected scale : 0.85, selected clustering : C_p

miserables: $|V|=77$, $|E|=254$ (Visible Nodes: $|V_N|=77$)

Description Scales: 0.852 | 0.936 | 0.952 | 0.85

Scores: $[C_p(\alpha=0.8500)] = [10: P=0.91, R=0.86, F0.50=0.45, F0.8500=0.76, \text{OMEGA}=9]$ $[C_0(\alpha=0.8500)] = [10: P=0.30, R=1.00, F0.50=0.46, F0.8500=0.89, \text{OMEGA}=0]$



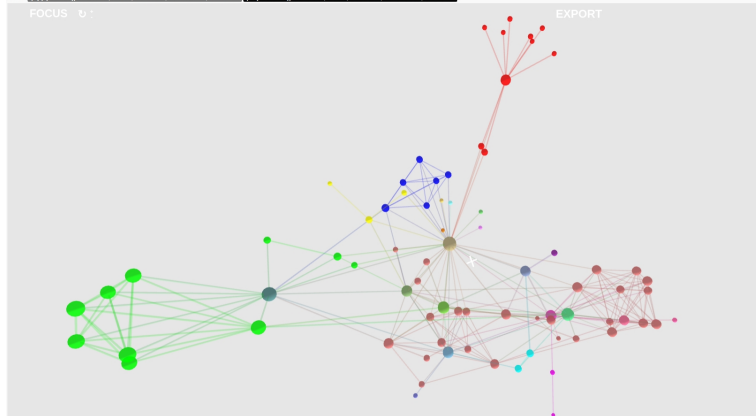
Informations on the graph and its clusterings

EXAMPLE : selected scale : 0.85, selected clustering : Co

miserables: $|V|=77$, $|E|=254$ (Visible Nodes: $|V_N|=77$)

Description Scales: 0.858 | 0.936 | 0.952 | 0.85

Scores: $[Co(su)=0.8500][c=10: P=0.91, R=0.86, F0.50=0.45, F0.8500=0.76, OMEGA=9]$ $[Co(sa)=0.8500][c=10: P=0.30, R=1.00, F0.50=0.46, F0.8500=0.89, OMEGA=0]$



Informations on the graph and its clusterings

EXAMPLE : selected scale : 0.916, selected clustering : C_p

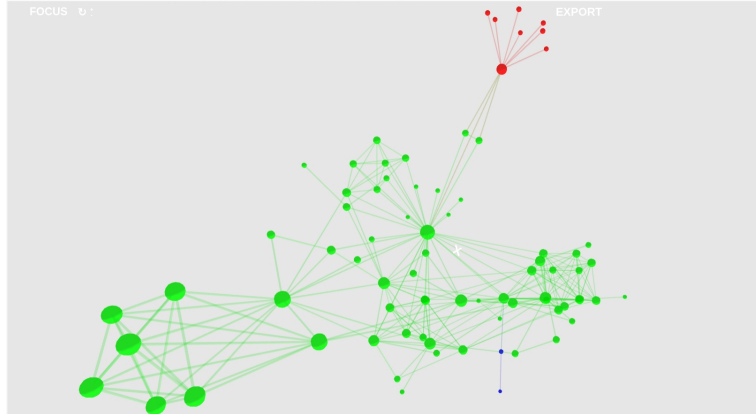
miserables: $|V|=77$, $|E|=254$ (Visible Nodes: $|V_N|=77$)

Description Scales: 0.916 | 0.936 | 0.952 | 0.969

Scores: $[C_p(\alpha)=0.9160][\alpha=0.916, R=0.98, F0.50=0.29, F0.9166=0.87, \text{OMEGA}=0]$ $[C_0(\alpha)=0.9160][\alpha=0.916, R=0.98, F0.50=0.29, F0.9166=0.88, \text{OMEGA}=0]$

FOCUS u :

EXPORT



Informations on the graph and its clusterings

EXAMPLE : selected scale : 0.916, selected clustering : C_0

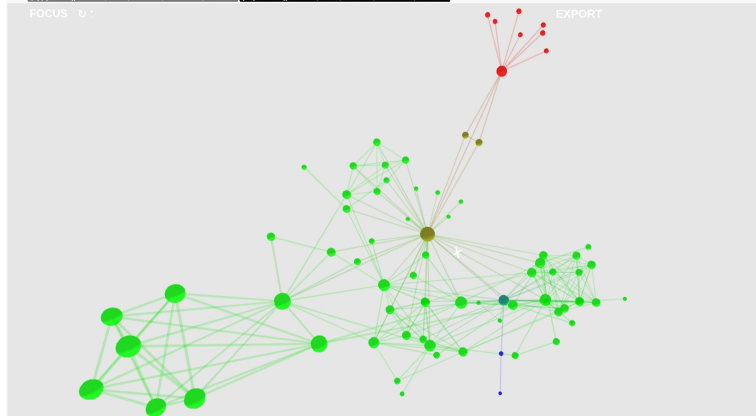
miserables: $|V|=77$, $|E|=254$ (Visible Nodes: $|V_N|=77$)

Description Scales: 0.916 | 0.936 | 0.952 | 0.960

Scores: $[C_0(s_0=0.916)] = 3$; $P=0.11$, $R=0.48$, $FD.50=0.29$, $FD.9166=0.87$, $OMEGA=0$ $[C_0(s_0=0.916)] = 3$; $P=0.11$, $R=1.00$, $FD.50=0.20$, $FD.9166=0.88$, $OMEGA=0$

FOCUS $\cup$:

EXPORT



Informations on a node

Hovering over a node : displays its modules and neighbors

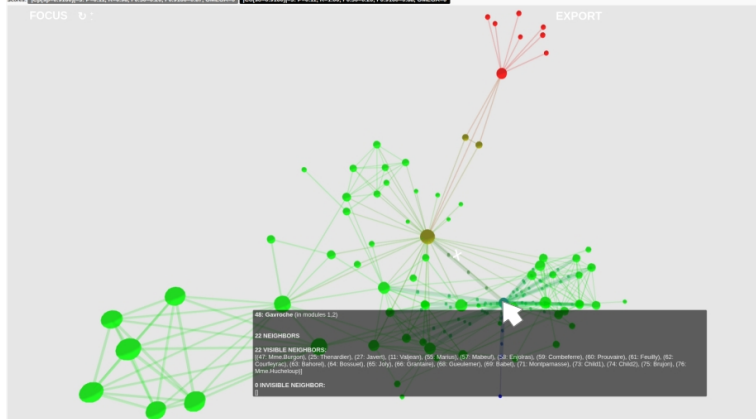
miserables: $|V|=77$, $|E|=254$ (Visible Nodes: $|V_N|=77$)

Description Scores: 0.918 | 0.936 | 0.92 | 0.89

Scores: $\langle C_{\text{ego}}=0.9169 \rangle=3$, $P=0.11$, $R=0.46$, $F0.50=0.26$, $F0.9169=0.87$, $\text{OMEGA}=0$ $\langle C_{\text{ego}}=0.9169 \rangle=3$, $P=0.11$, $R=1.00$, $F0.50=0.20$, $F0.9169=0.88$, $\text{OMEGA}=0$

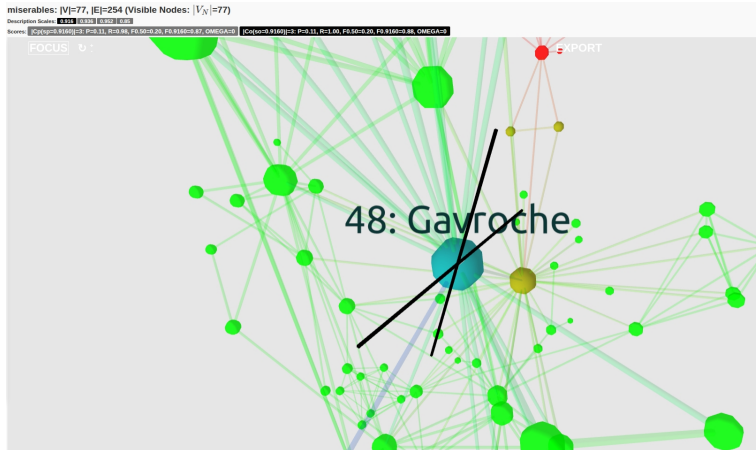
FOCUS u :

EXPORT



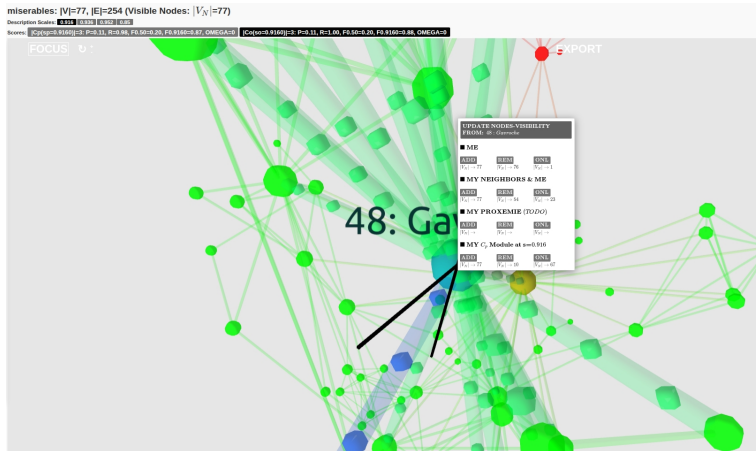
Informations on a node

Left-click on the node : zoom on it



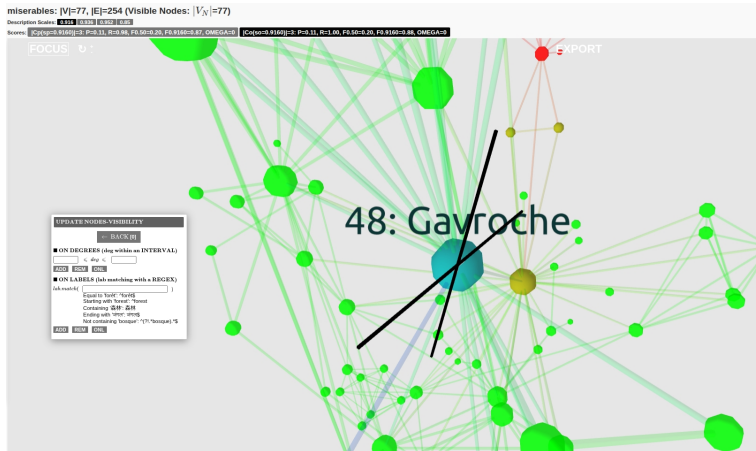
Selects Nodes-Visibility From one node

Rigth-click on the node : Update nodes-visibility from this node



Update nodes-visibility From on their properties

Right-click on empty space : Update nodes-visibility from their properties



FOCUS Button

Right-click on FOCUS button : Search for a node by its label or index

